

Major Project – Brief Report

Project Title:

Agentic Multimodal Retrieval for Long-Horizon Spatiotemporal Event Grounding

Abstract:

Existing surveillance retrieval systems are not suitable for long-horizon video as such models based on spatiotemporal text-to-video similarity are not effective for answering complex spatiotemporal queries. This project proposes an agentic video intelligence framework that combines deterministic computer vision tracking and generative multi-modal reasoning, all for production.

The ingestion engine ingests continuous video feeds at a desired frame rate and provides spatiotemporal tracks that preserve the identity of each frame of raw video using a fine-tuned YOLOv8 model and ByteTrack/DeepSORT. The tracks are structurally captioned with BLIP-2 and converted into multi-modal embeddings with CLIP. In the PostgreSQL backend, metadata is indexed, and the high-dimensional vectors are stored in the ChromaDB/Milvus backend.

The unstructured text, image or multi-lingual audio query (via Whisper) is passed to an autonomous LangGraph orchestrator for retrieval. The agent breaks down the query, performs “cross camera” semantic query search, and evaluates “temporal SQL” filtering. Using FFmpeg, the related video segments are sliced and immediately served to the Web and Flutter applications which are built by Next.js. The architecture substantially reduces the the overhead of the search space for the computation while achieving the temporal localization accuracy measured by Temporal Intersection over Union (t-IoU) on ForensicQA and UCF-Crime datasets.

Keywords:

Spatiotemporal Grounding, Agentic Retrieval, Multi-Object Tracking, VideoLLM, Long-Horizon Surveillance, Multimodal Embeddings.

Team Members:

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Scope of the Project:

This end-to-end AI Video Intelligence Platform works with 24/7 CCTV streams by RTSP and historical recordings, with the retention time set on the CCTV (5 days/14 days TTL). It includes auto frame extraction, captioning, multi-lingual audio search using Whisper, and natural language query decomposition. Plus, it provides advanced temporal bounded querying, enabling operators to narrow time and date range of incidents to highly specific areas, and a direct secure download module for all generated forensic clips. It is containerized using Docker and uses the frameworks FastAPI, React/Next.js and Flutter, scaling seamlessly for real-time security and criminal forensic operations.

Research Gaps (Maximum 4 to be Listed):

1. Existing video retrieval systems focus on analysing individual chunks of the video without taking into account the temporal relationships between them, which are important for a correct forensic analysis.
2. Current VideoRAG pipelines heavily depend on the basic similarity between text and video with CLIP and are incapable of tackling the complex spatiotemporal queries.
3. These systems do not have deep temporal grounding, and are not able to handle interleaving of image and text queries.
4. However, generative multi-modal models need to process dense, long duration video tensors with ample VRAM and compute to process the video, which makes the need for tracking-based pre-filtering a pressing business.

Research Objectives (Maximum 4 to be Listed):

1. To build a multi stage architecture that makes full use of a deterministic tracking algorithm to heavily prune the video search space and thus provide a plug and play architecture.
2. To divide long videos into actionable clips – person-centred or vehicle-centred – and drastically reduce the size of the indexable video database.
3. To integrate deterministic computer vision tracking pipelines with VideoL-LMs (like Qwen2-VL) for accurate temporal localization.
4. To precisely record the precise moments at which an event happens, and trace a suspect's path from one camera to another.

Brief Overview of the Proposed Method and Implementation:

Data Ingestion Pipeline: Raw videos and live CCTV RTSP streams are played at a given frame rate. They are fed into a carefully designed YOLOv8 model and then ByteTrack/DeepSORT is used to obtain continuous spatiotemporal coordinates of entities. Isolated targets are then fed into BLIP-2 for context-rich text captioning task and CLIP for high-dimensional multi-modal vector generation task. Last but not least, embeddings are

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stored within ChromaDB/Milvus, and structural tracking metadata, timelines and storage paths are stored in a PostgreSQL relational database.

Agentic Retrieval Pipeline: An unstructured text, multi-lingual audio (via Whisper), or interleaved image query is submitted by an operator, and the autonomous LangGraph orchestrator is in control. The agent breaks down the query into parts, looks in the vector space for semantically similar targets and uses SQL to filter time windows. The actual times are then mapped against the videos and dynamically generated to create relevant video clips and these are then served to Next.js Web Console and Flutter Mobile UI.

Key Technologies to be Used in the Implementation:

- **Vision & Tracking:** Python, PyTorch, OpenCV, YOLOv8, ByteTrack / DeepSORT.
- **Multimodal AI:** BLIP-2, CLIP, Qwen2-VL / VideoLLaMA, Whisper.
- **Orchestration:** LangGraph, LangChain, Celery.
- **Databases:** PostgreSQL (Metadata & TTL), ChromaDB / Milvus (Vector Engine).
- **Backend & DevOps:** FastAPI, AWS S3, Docker, Docker-Compose.
- **Frontend Ecosystem:** React / Next.js Web, TailwindCSS, Flutter Mobile SDK.

Team Declaration:

- All the Students in Team Have Been Informed About the Guidelines For the Major Project.
- All the Students Are Also Aware that the Project Carried Out During the IV Year, Shall be Submitted to a Conference or Journal On or Before 31-07-2026 and Project Shall be Completed in All Aspects by 31-08-2026.
- Students in the Team Will Adhere to the Timelines of the Major Project Without Fail.
- Plagiarism/Similarity Index for The Final Documentation Shall Be Less than or Equal to 07% and AI Tools Similarity Shall Also be Less Than or Equal to 10%.

Signature of the Co-Supervisor

Signature of the Supervisor

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Department Project Review Committee:

Name of Faculty	Designation / Role	Signature
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